Ebook

For Learners and Explorers

Robotics



Robotics and Edge AI

Innovate With NVIDIA Robotics

The NVIDIA three-computer solution enables robots to see, learn, perceive their surroundings, and make decisions in real time. This is driven by advances in AI, accelerated computing, robotic foundation models, physically accurate simulation, and a strong ecosystem of robotics partners.

Use Cases

Humanoid Robots—General-purpose humanoid robots are designed to quickly adapt to human-centric urban and industrial workspaces, tackling tedious, repetitive, or physically demanding tasks. They're increasingly being used in factories and healthcare facilities to assist humans and alleviate labor shortages through automation.

Synthetic Data Generation—Synthetic data addresses the challenge of data scarcity by providing text, videos, and images that can be used alongside real data to train multimodal physical AI models, saving time and reducing costs.

Industrial Facility Digital Twins—Virtual facilities, including factories, warehouses, distribution centers, semiconductor fabs, and data centers, unlock new possibilities for heavy industries. These virtual environments let you design, simulate, operate, and optimize assets and processes entirely in a digital space.

Robotics Simulation—Physical AI-powered robots need to autonomously perform complex tasks in dynamic environments. A "sim-first" approach is essential, allowing developers to train and validate these robots in physics-based digital environments before deployment.

Robot Learning—Preprogrammed robots struggle with unexpected changes, while AI-driven robots use simulation-based learning to adapt to dynamic environments. This lets them refine capabilities like navigation and manipulation, improving performance in a wide variety of scenarios.

Visual Resources

NVIDIA Robotics—The playlist includes videos that showcase the use of NVIDIA Jetson Orin™ developer kits, which are powerful platforms designed for AI at the edge. Overall, this playlist is a great resource for anyone interested in learning about the latest developments in the field of robotics and AI, whether you're a beginner or an experienced developer.



The resources on this page will link you to the self-paced courses relevant to specific topics. The point of emphasis in these workbooks is to highlight the prerequisites you need to dive into the offerings from NVIDIA.

Topics	Courses
Robotics Foundations (Free)	**A Beginner's Guide to Autonomous Robots** No prior knowledge of robotics is required, but a general understanding of software frameworks and programming concepts is beneficial.
Synthetic Data Generation (Cost)	**An Introduction to NVIDIA Cosmos™ for Physical AI** An understanding of AI model training, fine-tuning, and deep learning frameworks; Experience with Python programming; Basic experience with 3D software and workflows.
	Generating High-Quality Motion Data for Robotics With MobilityGen Basic understanding of robotics and simulations; NVIDIA Isaac Sim™ 4.5 and corresponding technical specifications.
Isaac ROS (Cost)	**An Introduction to AI-Based Robot Development With Isaac ROS** No prerequisites are required to take this course.
	Accelerating ROS 2 With NVIDIA GPU-Powered Libraries and AI Models Experience with OpenUSD, NVIDIA Omniverse™ and 3D CAD software; Experience in Python; Basic knowledge of NVIDIA Isaac Sim and Isaac ROS. A basic understanding of 3D applications is useful, but not required. You must have VS Code and Git installed before you begin this course.
Isaac Sim (Free)	**Getting Started: Simulating Your First Robot in Isaac Sim** Basic Python knowledge and familiarity with robotics concepts. A Linux machine meeting Isaac Sim's system requirements is necessary for this course and for properly running simulations.

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Topics	Courses
Isaac Sim (Free) *continued*	**Ingesting Robot Assets and Simulating Your Robot in Isaac Sim** This is the second course in the *Getting Started With Isaac Sim* learning path. Please complete *Getting Started: Simulating Your First Robot in Isaac Sim* before beginning this course. Basic Python knowledge and familiarity with robotics concepts. A Linux machine meeting Isaac Sim's system requirements is necessary for this course and for properly running simulations.
	Synthetic Data Generation for Perception Model Training in Isaac Sim This is the third course in the *Getting Started With Isaac Sim* learning path. Please complete *Ingesting Robot Assets and Simulating Your Robot in Isaac Sim* before beginning this course. Basic Python knowledge and familiarity with robotics concepts. A Linux machine meeting Isaac Sim's system requirements is necessary for this course and for properly running simulations.
	Developing Robots With Software-in-the-Loop (SIL) In Isaac Sim This is the fourth course in the *Getting Started With Isaac Sim* learning path. Please complete *Synthetic Data Generation for Perception Model Training in Isaac Sim* before beginning this course. Basic Python knowledge and familiarity with robotics concepts. A Linux machine meeting Isaac Sim's system requirements is necessary for this course and for properly running simulations.

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Topics	Courses
Isaac Sim (Free) *continued*	**Leveraging ROS 2 and Hardware-in-the-Loop (HIL) in Isaac Sim** This is the fifth course in the *Getting Started With Isaac Sim* learning path. Please complete *Developing Robots with Software-in-the-Loop (SIL) In Isaac Sim* before beginning this course. Basic Python knowledge and familiarity with robotics concepts. A Linux machine meeting Isaac Sim's system requirements is necessary for this course and for properly running simulations.
	Software-in-the-Loop Testing for Robots With OpenUSD, Isaac Sim, and ROS Experience with OpenUSD, NVIDIA Omniverse, and 3D CAD software. Experience in Python. Basic knowledge of NVIDIA Isaac Sim and Isaac ROS.
	Develop, Simulate, and Deploy Robot Intelligence With Scaled Foundations Basic knowledge of NVIDIA Isaac Sim, robotics fundamentals, and Python.
Isaac Lab (Free)	**An Introduction to Robot Learning and Isaac Lab** No prior knowledge of robotics is required, but a general understanding of software frameworks and programming concepts is beneficial.
	Transferring Robot Learning Policies From Simulation to Reality This course does not have a hands-on component. No prerequisites are required to take this course.

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Topics	Courses
Isaac Lab (Free) *continued*	**Train Your First Robot in Isaac Lab** Basic familiarity with Python, the command line, and robotics concepts.
	Train Your Second Robot in Isaac Lab Basic familiarity with Python, the command line, and robotics concepts.
Jetson Nano (Cost)	**Getting Started with AI on NVIDIA Jetson Nano™** Basic familiarity with Python (helpful, not required). **Related Training:** You may be interested in the following free self-paced training on NVIDIA Jetson™: **>** Jetson AI Lab
	Getting Started With AI on Jetson Nano (instructor-led)

Explore Additional Resources

NVIDIA On-Demand

The NVIDIA online video library offers a wide range of technical lessons and lectures from the NVIDIA GTC technology conference and other industry-leading events. You'll also find podcasts, demos, research articles, and learning videos on AI, climate science, language processing, autonomous machines, design, Omniverse, and more.

Explore content for:

- > NVIDIA Training Labs On-Demand
- > Generative AI
- > Deep learning
- > Researchers and educators

NVIDIA Webinars

Webinars let students and educators dive deeper into specific topics and discover innovations in AI, deep learning, robotics, autonomous machines, data science, and more.

- > AI For All: Educators and Students
- > What's New With NVIDIA Certification
- > How to Accelerate Your Career in AI
- > Accelerating Gene Variant Detection With Deep Learning
- > Deploying Generative AI in Production
- > Empowering Future Engineers and Scientists With AI and NVIDIA Modulus
- > Essential Training and Tips to Accelerate Your Career in AI

Check out the full webinar catalog >

NVIDIA YouTube Page

Discover a wide range of videos on the NVIDIA YouTube page, covering the latest in AI technology, tutorials, and expert talks. Stay updated and enhance your knowledge by watching our curated content, including hands-on technical demos and insightful discussions.

Visit the YouTube Channel >

Connect on Social

Don't miss out on the conversations happening on our social platforms. Follow us to stay updated on the latest trends, insights, and events in AI education.

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Questions? Requests? Get in Touch.

We'd love the opportunity to support and collaborate with your school or organization. Contact us at: developer.nvidia.com/contact

To learn more how NVIDIA is helping to usher in the next era of computing in higher education and research, visit: nvidia.com/higher-education-research/

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